CS391R1: Computer & Memory Architectures
Lab 4: Load, S-type, and B-type

Due: Friday Nov 14, 2025 at 2:30pm ET (before Discussion Section)

NOTE: Both Part 1 and Part 2 are due Friday Nov 14.

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1. Part 1: Design Exercises (Individual Work)
    Problem 1.1: Virtual Memory and TLB
2. Part 2: Lab Activity (Individual Work)
    (Added!) Lab 3 Problem 2.3: BRAM as Instruction Memory
    Introduction: Load, S-type, and B-type
    Problem 2.1: I-type Load
    Problem 2.2: S-type Store
    Problem 2.3: B-type Branches
3. Submission Instructions
    For Part 1: Design Exercises
    For Part 2: Lab Activity
    Summary of Deliverables
Appendix 1: Loading Data into BRAM (Use Binary File on Blackboard)

NOTE: As announced, we have added Lab 3 Problem 2.3 to the Lab 4 handout. If you did not finish Lab 3 Problem 2.3 by its original deadline, then please finish it by the Lab 4 deadline (see above), and we will go back and retroactively add those points back to your Lab 3 score, saturating at the max points that Problem 2.3 was originally worth.

Changelog:

- Added Lab 3 Problem 2.3 to the Lab 4 handout
- Added Part 1 Design Exercise

1. Part 1: Design Exercises (Individual Work)

Problem 1.1: Virtual Memory and TLB

Consider a RISC-V processor that includes a 40-bit virtual address, an MMU that supports 4096 (**2^{12}**) bytes per page, **2^{32}** bytes of physical memory, and a large Flash memory that serves as a disk. The MMU and the page fault handler implement an LRU replacement strategy.

(a) What is the size of the page map for this processor? Assuming the page map includes the standard dirty and resident bits, specify the width of each page map entry in bits, and number of entries in the page map.

Size of page map entry in bits: 22 ______

Number of entries in the page map: 2^28 ______

(b) The following test program is running on this RISC-V processor:

Address	Instruction
`0x0`	`addi x3, x0, 0x5590`
`0x4`	`lw x5, 0(x3)`
`0x8`	`sw x5, 0x6200(x0)`

The first 8 locations of the page table, just before executing this test program, are shown below; the least-recently-used (“LRU”) and next least-recently-used page (“next LRU”) are as indicated:

Page Map

	VPN	D	R	PPN
	0	1	1	0x4
	1	0	1	0x5
	2	0	1	0x3
LRU →	3	1	1	0x8
	4	0	1	0x0
	5	–	0	–
	6	0	1	0x2
Next LRU →	7	0	1	0x6

This RISC-V processor also has a 4 element, fully associative Translation Lookaside Buffer (TLB) that caches page map translations from VPN to PPN:

TLB

	Tag (VPN)	D	R	PPN
LRU →	0x3	1	1	0x8
	0x2	0	1	0x3
	0x6	0	1	0x2
Next LRU →	0x1	0	1	0x5

For each **unique** virtual page that is accessed by this program, specify the VPN, whether or not it results in a TLB hit on the **first** access to that page, whether or not it results in a page fault, and the PPN that the page ultimately maps to. You may not need to use all rows of the table.

VPN	TLB Hit (Yes/No)	Page Fault (Yes/No)	PPN
0x0	no	no	0x4
0x5	no	yes	0x8
0x6	yes	no	0x2

(c) Which physical pages, if any, need to be written to disk during the execution of the test program in part (b)?

Physical page numbers written to disk or NONE: 0x8

(d) What is the physical address of the **`lw`** instruction?

Physical address of **`lw`** instruction: 0x 4004

NOTE: Lab 4 has a substantial implementation component in Part 2, so we will only have one written design exercise in Part 1. The Lab Report for Lab 4 Part 2 will be worth a large portion of the total points for this assignment, so please read the submission instructions carefully and leave yourself enough time to test your code, capture waveforms, and write your Lab Report.